

# Functional Specification Document

## ChurnGuard — Churn-Risk Detection & Retention Workflow

|             |                                         |
|-------------|-----------------------------------------|
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This document specifies how the functional requirements behave in build terms: the scoring algorithm, object and field definitions, screen behaviour, validation rules and error handling. Section references from the FRD and RTM point here.

### 1. Solution architecture summary

| Layer         | Component                  | Technology                                                                              | Responsibility                                                            |
|---------------|----------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Orchestration | Nightly scoring workflow   | n8n scheduled workflow                                                                  | Sequences extraction, scoring trigger, post-scoring sync; handles retries |
| Extraction    | Five signal extractors     | SQL against usage mart; REST for connector health; native reads for support and billing | Produce one staging row per account per signal                            |
| Scoring       | Scoring engine             | Deluge scheduled functions, chunked at 200 accounts per execution                       | Normalise, sub-score, weight, composite, tier, movement                   |
| Persistence   | Custom objects             | Zoho CRM custom modules                                                                 | Score history, signal detail, cases, tasks, config, audit                 |
| Automation    | Alert and SLA layer        | Deluge workflow rules + custom schedules                                                | Task creation, routing, SLA clock, breach detection                       |
| Presentation  | Risk panel                 | CRM Canvas / related list                                                               | Account-level explainability                                              |
| Analytics     | Retention Health dashboard | Zoho Analytics                                                                          | Portfolio reporting                                                       |

### 2. Nightly scoring cycle — sequence

| Step | Action                                                                                                                                             | On failure                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1    | Build in-scope account list: <code>Status = Active</code> AND <code>Segment = High-Value</code> AND no submitted cancellation (BRule-01, BRule-02) | Abort cycle, alert ops, retain prior scores                               |
| 2    | Load active configuration version; validate weights total 100%                                                                                     | Abort; alert Business Administrator                                       |
| 3    | Extract S1-S5 staging data in parallel                                                                                                             | Per-interface retry ×3 at 15 min; then mark that signal Insufficient Data |
| 4    | Apply data-quality evaluation per account per signal (FR-007)                                                                                      | —                                                                         |
| 5    | Apply suppression or weight redistribution (FR-008, FR-009)                                                                                        | —                                                                         |
| 6    | Calculate sub-scores, weighted contributions, composite (FR-010, FR-011)                                                                           | Log and skip account; add to exception report                             |
| 7    | Determine tier with hysteresis, compare to previous tier (FR-016 – FR-018)                                                                         | —                                                                         |
| 8    | Write <code>Risk_Score_History</code> and five <code>Signal_Detail</code> rows per account (FR-019)                                                | Transaction rolled back per account; account listed in exceptions         |
| 9    | Generate alerts and tasks for escalations, unless shadow run active (FR-021 – FR-027, FR-030)                                                      | Retry queue; unsent alerts reported                                       |
| 10   | Trigger analytics sync (IF-06)                                                                                                                     | Dashboard shows stale-refresh warning                                     |
| 11   | Write run log: counts scored, suppressed, reduced, exceptions, duration, config version (NFR-22)                                                   | —                                                                         |

**Idempotency (NFR-08):** the engine writes history keyed on `Account_ID + Score_Date`. A re-run for the same date updates the existing row rather than inserting a duplicate, and appends a re-run note to the run log.

### 3. Scoring algorithm

#### 3.1 Composite calculation

```

sub_score(signal)      =  $\Sigma$  (component_points) capped at 100
weighted(signal)      = sub_score  $\times$  weight
composite              = ROUND(  $\Sigma$  weighted(signal) / 100 )  $\rightarrow$  integer 0–100
dominant_driver        = signal with MAX(weighted)

```

Ties on dominant driver resolve by the fixed precedence S1 > S2 > S3 > S4 > S5.

### 3.2 Signal S1 — Product usage decline • weight 30%

| Component                   | Measure                                                     | Band                     | Points |
|-----------------------------|-------------------------------------------------------------|--------------------------|--------|
| Active-day decline          | active_days_30d vs 30-day equivalent of avg_active_days_90d | $\geq 60\%$ decline      | 45     |
|                             |                                                             | 40–59% decline           | 30     |
|                             |                                                             | 20–39% decline           | 15     |
|                             |                                                             | < 20% decline or growth  | 0      |
| Order volume decline        | orders_processed_30d vs 90d average                         | $\geq 50\%$ decline      | 30     |
|                             |                                                             | 25–49% decline           | 18     |
|                             |                                                             | < 25% decline            | 0      |
| Feature breadth contraction | distinct_modules_30d vs distinct_modules_prior_90d          | Dropped $\geq 2$ modules | 15     |
|                             |                                                             | Dropped 1 module         | 8      |
|                             |                                                             | No change or increase    | 0      |
| Dormancy                    | Days since last login                                       | $\geq 21$ days           | 25     |
|                             |                                                             | 14–20 days               | 12     |
|                             |                                                             | < 14 days                | 0      |

### 3.3 Signal S2 — Marketplace & store health • weight 25%

| Component                   | Measure                             | Band                             | Points |
|-----------------------------|-------------------------------------|----------------------------------|--------|
| Connector disconnect        | Any connector in Disconnected state | $\geq 7$ days                    | 40     |
|                             |                                     | 3–6 days                         | 25     |
|                             |                                     | < 3 days                         | 10     |
| Sync error rate             | sync_errors_30d / sync_attempts_30d | > 10%                            | 30     |
|                             |                                     | 5–10%                            | 18     |
|                             |                                     | < 5%                             | 0      |
| GMV decline                 | gmv_30d vs 90d average              | $\geq 40\%$ decline              | 25     |
|                             |                                     | 20–39% decline                   | 12     |
|                             |                                     | < 20% decline                    | 0      |
| Marketplace account warning | Health flag from marketplace        | Suspension or restriction active | 30     |
|                             |                                     | Warning active                   | 15     |
|                             |                                     | None                             | 0      |

Seasonality note agreed with CS leadership: GMV decline points are suppressed for accounts whose primary marketplace category is flagged seasonal during a configured seasonal window, to avoid a January false-positive wave across the seller base.

### 3.4 Signal S3 — Support experience • weight 20%

| Component                    | Measure                            | Band                       | Points |
|------------------------------|------------------------------------|----------------------------|--------|
| Ticket volume above baseline | tickets_30d vs 90d monthly average | $\geq 2\times$ baseline    | 30     |
|                              |                                    | 1.5–1.99 $\times$ baseline | 18     |
|                              |                                    | < 1.5 $\times$             | 0      |
| Severity                     | Count of P1 tickets in 30d         | $\geq 2$                   | 25     |
|                              |                                    | 1                          | 12     |
| SLA breaches                 | Breached tickets in 30d            | $\geq 2$                   | 20     |
|                              |                                    | 1                          | 10     |
| Reopened tickets             | Reopens in 30d                     | $\geq 2$                   | 15     |

|              |                        |     |    |
|--------------|------------------------|-----|----|
| Satisfaction | Most recent CSAT (1–5) | ≤ 2 | 25 |
|              |                        | 3   | 12 |
|              |                        | ≥ 4 | 0  |

### 3.5 Signal S4 — Commercial & billing • weight 15%

| Component                          | Measure                                     | Band                 | Points |
|------------------------------------|---------------------------------------------|----------------------|--------|
| Failed payments                    | Attempts failed in 60d                      | ≥ 2                  | 35     |
|                                    |                                             | 1                    | 18     |
| Invoice ageing                     | Oldest overdue invoice                      | > 60 days            | 30     |
|                                    |                                             | 31–60 days           | 18     |
|                                    |                                             | ≤ 30 days            | 0      |
| Contraction                        | Plan or seat change vs prior                | Plan downgrade       | 30     |
|                                    |                                             | Seat reduction ≥ 25% | 20     |
| Renewal proximity without activity | Days to renewal, no logged renewal activity | ≤ 60 days            | 25     |
|                                    |                                             | 61–90 days           | 12     |

### 3.6 Signal S5 — Relationship engagement • weight 10%

| Component       | Measure                                    | Band                        | Points |
|-----------------|--------------------------------------------|-----------------------------|--------|
| Contact silence | Days since last meaningful logged activity | ≥ 90                        | 40     |
|                 |                                            | 60–89                       | 25     |
|                 |                                            | 30–59                       | 10     |
| QBR attendance  | QBRs attended in trailing two quarters     | 0 of 2                      | 25     |
|                 |                                            | 1 of 2                      | 12     |
| Champion status | Primary contact record                     | Marked inactive or departed | 30     |
| Responsiveness  | Response rate to outbound, 90d             | < 20%                       | 20     |
|                 |                                            | 20–49%                      | 10     |

### 3.7 Tiering and hysteresis

| Tier     | Composite score |
|----------|-----------------|
| Stable   | 0 – 39          |
| Watch    | 40 – 69         |
| Critical | 70 – 100        |

Hysteresis (FR-018): an account already in a tier does not move until the score crosses the boundary by more than the configured buffer (default 3 points). An account at Watch with a score of 38 stays at Watch; at 36 it moves to Stable. This prevents alert churn on accounts oscillating at a boundary.

### 3.8 Worked example — account CG-DEMO-114

| Signal         | Key inputs                                                                                   | Sub-score              | Weight | Weighted  |
|----------------|----------------------------------------------------------------------------------------------|------------------------|--------|-----------|
| S1 Usage       | Active days 14 vs 38 avg (–63%); orders –31%; modules –1; last login 9d                      | 45 + 18 + 8 + 0 = 71   | 30%    | 21.3      |
| S2 Marketplace | Connector disconnected 4 days; sync errors 7%; GMV –22%; no warning                          | 25 + 18 + 12 + 0 = 55  | 25%    | 13.8      |
| S3 Support     | Tickets 1.6× baseline; 1 P1; 1 SLA breach; CSAT 3                                            | 18 + 12 + 10 + 12 = 52 | 20%    | 10.4      |
| S4 Commercial  | No failed payments; invoice 34 days overdue; no contraction; renewal in 74 days, no activity | 0 + 18 + 0 + 12 = 30   | 15%    | 4.5       |
| S5 Engagement  | Last activity 67 days; 1 of 2 QBRs; champion active; response rate 41%                       | 25 + 12 + 0 + 10 = 47  | 10%    | 4.7       |
| Composite      |                                                                                              |                        |        | 54.7 → 55 |

Result: tier **Watch**, previous tier Stable → **Escalation**. Dominant driver **S1 Product usage decline** (21.3 weighted points). Playbook attached: *Usage Recovery — adoption review*. Task due in 3 business days.

## 4. Object and field specification

### 4.1 Risk\_Score\_History (custom module)

|  |
|--|
|  |
|--|

| Field             | Type                    | Length / values                      | Mandatory | Default        | Validation                                    |
|-------------------|-------------------------|--------------------------------------|-----------|----------------|-----------------------------------------------|
| Score_Date        | Date                    |                                      | Y         | Cycle date     | Unique with Account_ID                        |
| Account           | Lookup → Account        |                                      | Y         |                | Must be in-scope at cycle time                |
| Composite_Score   | Integer                 | 0-100                                | N         |                | Null only when Data_Quality_Flag = Suppressed |
| Tier              | Picklist                | Stable, Watch, Critical              | N         |                | Null when suppressed                          |
| Previous_Tier     | Picklist                | Stable, Watch, Critical              | N         |                | Null on first cycle                           |
| Movement          | Picklist                | No Change, Escalation, De-escalation | Y         | No Change      |                                               |
| Dominant_Driver   | Picklist                | S1...S5                              | N         |                |                                               |
| Data_Quality_Flag | Picklist                | Complete, Reduced, Suppressed        | Y         | Complete       |                                               |
| Reduced_Signals   | Multi-select            | S1...S5                              | N         |                | Populated when flag = Reduced                 |
| Config_Version    | Lookup → Config_Version |                                      | Y         | Active version |                                               |
| Override_Active   | Checkbox                |                                      | Y         | False          |                                               |

Record-level: read-only to all application roles after creation (DQ-05, NFR-14).

#### 4.2 Signal\_Detail (child of Risk\_Score\_History)

| Field                 | Type                   | Notes                                           |
|-----------------------|------------------------|-------------------------------------------------|
| Signal_Code           | Picklist S1-S5         | Mandatory                                       |
| Sub_Score             | Integer 0-100          | Mandatory unless insufficient                   |
| Weight_Applied        | Decimal (5,2)          | Post-redistribution value                       |
| Weighted_Contribution | Decimal (5,2)          |                                                 |
| Raw_Inputs            | Multi-line text (JSON) | Each component with value and comparison period |
| Comparison_Period     | Text                   | e.g. "30d vs 90d avg to 2026-02-28"             |
| Insufficient_Data     | Checkbox               |                                                 |

#### 4.3 Retention\_Case

| Field                        | Type                           | Values / rules                                                                            |
|------------------------------|--------------------------------|-------------------------------------------------------------------------------------------|
| Case_Number                  | Auto-number                    | RC - {00000}                                                                              |
| Account                      | Lookup                         | Mandatory; only one open case per account (FR-039)                                        |
| Opening_Score / Opening_Tier | Integer / Picklist             | Copied at creation, never updated                                                         |
| Dominant_Driver              | Picklist                       | Copied at creation                                                                        |
| Playbook_Applied             | Lookup → Playbook              | Defaulted from driver mapping; editable                                                   |
| Owner                        | User lookup                    | Defaults to account's owning CSM                                                          |
| Status                       | Picklist                       | Open, In Progress, Awaiting Customer, Closed-Saved, Closed-Churned, Closed-False Positive |
| Actions_Taken                | Related list → Case_Action_Log |                                                                                           |
| Closure_Outcome              | Picklist                       | Mandatory when Status starts with "Closed"                                                |
| Closure_Reason               | Picklist                       | Controlled list; mandatory on closure (FR-038)                                            |
| False_Positive_Signal        | Picklist S1-S5                 | Mandatory when Status = Closed-False Positive (BRule-07)                                  |
| Renewal_Result               | Formula / lookup               | Pulled from linked renewal record (FR-041)                                                |
| Manual_Hold                  | Checkbox                       | Blocks auto-closure (FR-040)                                                              |

#### 4.4 Retention\_Task

| Field                  | Type        | Notes                                                       |
|------------------------|-------------|-------------------------------------------------------------|
| Account, Case          | Lookups     | Case may be null until CSM opens one                        |
| Assignee               | User lookup | Owning CSM, else CS Manager queue (FR-025)                  |
| Created_At             | DateTime    | System                                                      |
| SLA_Tier               | Picklist    | Critical / Watch                                            |
| Due_Date               | Date        | Created_At + SLA business days on CS calendar (BRule-05)    |
| First_Touch_At         | DateTime    | Set by automation on first logged activity after Created_At |
| Elapsed_Business_Hours | Formula     | Created_At → First_Touch_At                                 |
| SLA_Status             | Picklist    | Within SLA / Breached / Met                                 |

4.5 Config\_Version

Effective\_Date, Weights (S1-S5), Tier boundaries, Hysteresis\_Buffer, SLA\_Critical\_Days, SLA\_Watch\_Days, Playbook\_Mapping (related list), Changed\_By, Prior\_Version, Change\_Note. Weights validated to total exactly 100% on save (FR-048).

5. Screen behaviour

5.1 Account Risk Panel (see wireframe W-01)

| Element             | Behaviour                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score dial          | Shows composite, tier label and tier icon. Colour is supported by shape and text (NFR-17).                                                                                           |
| Movement indicator  | Arrow with delta vs previous cycle and days in current tier.                                                                                                                         |
| Signal breakdown    | Five rows, sorted by weighted contribution descending. Each row: signal name, sub-score, contribution bar, primary driving value in plain language ("Active days 14 vs 38 average"). |
| Row expand          | Reveals every component with value, band matched and points awarded.                                                                                                                 |
| Data quality banner | Shown when flag is Reduced or Suppressed; names the unavailable signals (FR-034).                                                                                                    |
| Trend               | 90-day sparkline of composite score with tier bands shaded.                                                                                                                          |
| Actions             | <i>Open retention case</i> (pre-populated), <i>Rescore now</i> , <i>Override tier</i> .                                                                                              |
| Empty state         | Where the account is out of scope: "Not scored — outside the high-value segment" with the segment rule named.                                                                        |

5.2 Retention Task consumption

Task opens with the alert summary at the top: score, movement, dominant driver, top three signals, playbook link, and a deep link to the risk panel. *Start case* converts the task into an in-progress Retention Case and stamps first touch.

5.3 Retention Health Dashboard (see wireframe W-03)

| Tile              | Definition                                                                |
|-------------------|---------------------------------------------------------------------------|
| Accounts by tier  | Count and % of in-scope accounts per tier, current cycle                  |
| ARR by tier       | Sum of account ARR per tier                                               |
| Tier movement     | Escalations and de-escalations in the selected period                     |
| SLA compliance    | Tasks with first touch on or before due date ÷ tasks due, in period       |
| Save rate         | Closed-Saved ÷ (Closed-Saved + Closed-Churned) for cases opened in period |
| Critical worklist | Account, CSM, score, days in tier, case status — sorted by ARR descending |
| Signal mix        | Distribution of dominant drivers across Critical accounts                 |

5.4 Configuration screen (see wireframe W-04)

Weights entered as integers; running total displayed live and blocks save unless it equals 100. *Preview impact* runs the proposed configuration against the last cycle's stored inputs and returns resulting tier counts and estimated daily alert volume, without saving (FR-052). Save requires a change note and creates a new Config\_Version effective from the next cycle.

6. Validation and error handling

| Ref  | Condition                                  | System behaviour                      | User message                                                                                                         |
|------|--------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| V-01 | Weights do not total 100%                  | Block save                            | "Weights must total 100%. Current total: {n}%."                                                                      |
| V-02 | Tier boundaries overlap or leave a gap     | Block save                            | "Tier ranges must be continuous and must not overlap. Check the Watch upper bound."                                  |
| V-03 | Case closure without reason                | Block save                            | "Select a closure reason before closing this case."                                                                  |
| V-04 | Closed-False Positive without signal named | Block save                            | "Select which signal produced the false reading."                                                                    |
| V-05 | Second open case on one account            | Block create, deep-link existing      | "This account already has an open retention case ({Case_Number}). Open it instead."                                  |
| V-06 | Override expiry more than 90 days ahead    | Block save                            | "An override can run for a maximum of 90 days. Choose a date on or before {date}."                                   |
| V-07 | Source feed stale > 48h                    | Mark signal insufficient, continue    | Panel banner: "{Signal} unavailable — data last received {date}. Score calculated on the remaining signals."         |
| V-08 | Two or more signals insufficient           | Suppress score, retain previous       | Panel banner: "Scoring paused for this account — {signals} unavailable since {date}. Showing the score from {date}." |
| V-09 | Scoring cycle fails twice                  | Retain prior scores, alert operations | Dashboard banner: "Scores last updated {timestamp}."                                                                 |

|      |                                                 |                                     |                                                                            |
|------|-------------------------------------------------|-------------------------------------|----------------------------------------------------------------------------|
| V-10 | Account has no active owning CSM                | Route to CS Manager queue, flag gap | Task note: "No account owner assigned — routed to {team} queue."           |
| V-11 | On-demand rescore requested while cycle running | Queue, do not run concurrently      | "Nightly scoring is running. Your rescore will complete when it finishes." |

Error messages state what happened and what to do next, in the interface's voice, without apology (per the CRM content standard).

## 7. Security model

| Profile                   | Score & signals      | Cases               | Config       | Audit log |
|---------------------------|----------------------|---------------------|--------------|-----------|
| CSM                       | Read — own accounts  | Create, edit own    | None         | None      |
| CS Manager                | Read — team accounts | Read all team, edit | None         | Read team |
| VP Customer Success       | Read all             | Read all            | Approve only | Read all  |
| Business Administrator    | Read all             | Read all            | Create, edit | Read all  |
| Data & BI Analyst         | Read all             | Read all            | None         | Read all  |
| System / integration user | Create, update       | Create              | Read         | Create    |

No profile has delete rights on Risk\_Score\_History, Signal\_Detail or Audit\_Entry.

## 8. Migration and cutover

| Step | Detail                                                                                                                          |
|------|---------------------------------------------------------------------------------------------------------------------------------|
| M-1  | Backfill 90 days of score history from historical source data so trends and baselines exist at go-live                          |
| M-2  | Load playbook content and driver mapping from CS Enablement                                                                     |
| M-3  | Load CS working calendar including regional public holidays (open item O-1)                                                     |
| M-4  | Run 14-day shadow period: scores and history written, alerts suppressed (FR-030)                                                |
| M-5  | Calibration review — compare shadow alert volume and precision against the historical sample; tune thresholds via configuration |
| M-6  | Enable alerting; capture BO baselines in the preceding week                                                                     |